Human Computer Interaction

Dr. Sevgi Koyuncu Tunç

“**H**uman **C**omputer **I**nteraction” concept has been introduced to understand and design the **relationship** between **humans** and **computers**.

A good **user interface** does not come out by accident; it needs to benefit from Human - Computer Interaction discipline.

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SENG 474 Human Computer Interaction Course Outline

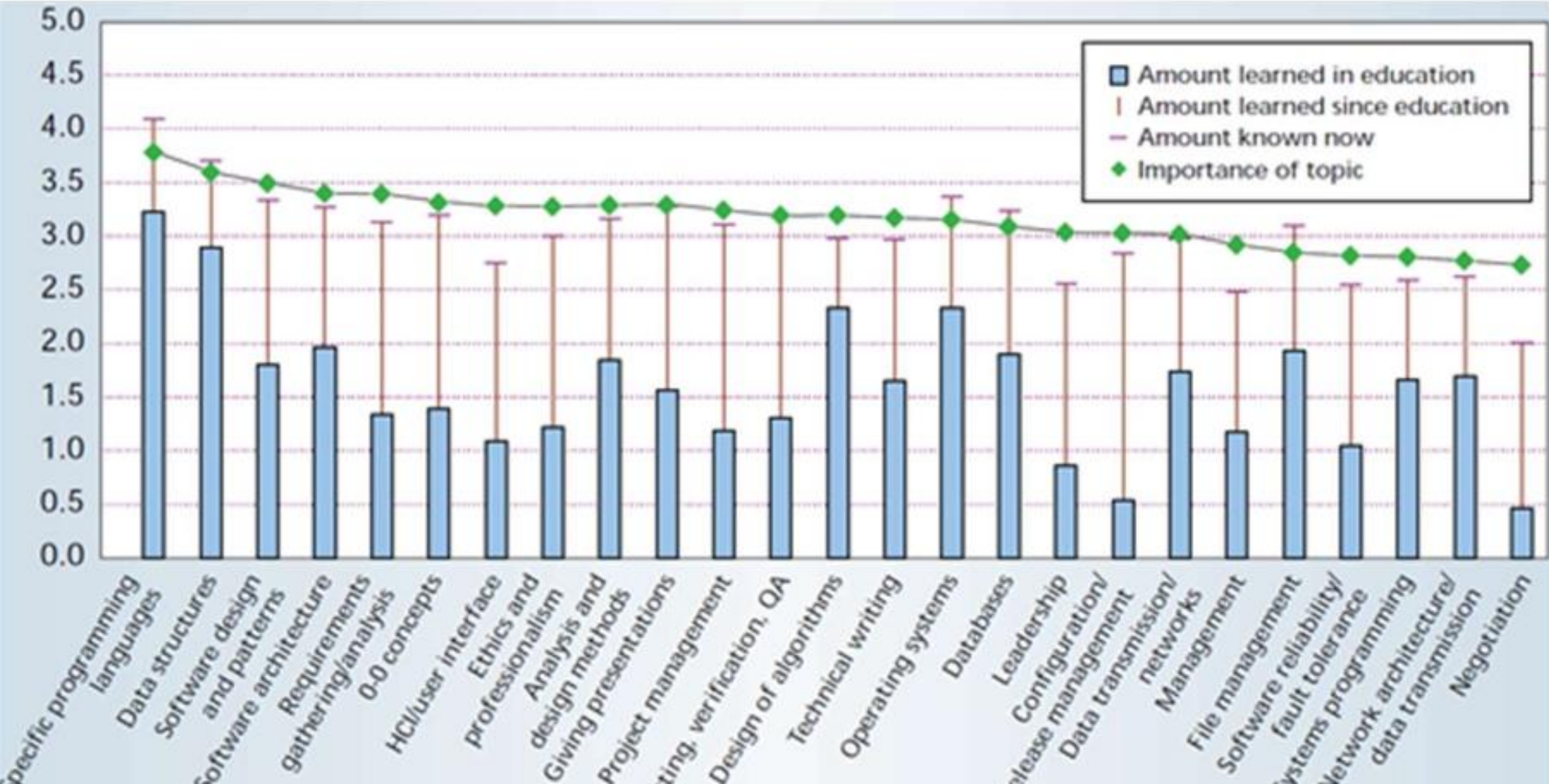
1. Introduction, history of HCI
2. Affordance, usability. Usability principles.
3. Human information processing limitations, human cognitive and sensory limits.
4. Prototyping techniques
5. Evaluating usability. Predictive and interpretive evaluation.
6. User interface design paradigms, User interface design rules
7. Midterm Review
8. Process of interaction design. Design languages.
9. User modeling and the user profile. Adaptive interfaces.
10. 3D User interfaces, augmented reality.
11. Speech recognition and conversational interfaces.
12. Multimodal interfaces, perceptual interfaces. Ubiquitous Computing
13. Project Presentations
14. Project Presentations

What is HCI?

- “Human Computer Interaction” (HCI) is a concept used to understand and design the relationship between humans and computers.

- HCI principals should be deemed vital in the software development life cycle because benefits of usability include reduced maintenance costs and increased customer satisfaction.

What information is important for a software professional?



TR SW Engineers' HCI Knowledge

53.9% have not received an HCI training at all.

22% assess themselves competent in HCI.

71% need training in HCI

86% not familiar with natural user interfaces.

Using templates prepared by UX Designers is uncommon

Prototyping and customer approval on designs is not popular.

End user involvement is low in design and development phases of software life cycle.

Developers are not aware of new usability testing approaches.

2019 Turkey User Experience Report;

- 60% of the companies are not allocating sufficient budget for the user experience activities,
- 31% are not using any usability techniques,
- and there is a lack of knowledge on usability concepts (UXServices, 2019)

Booth - 1989

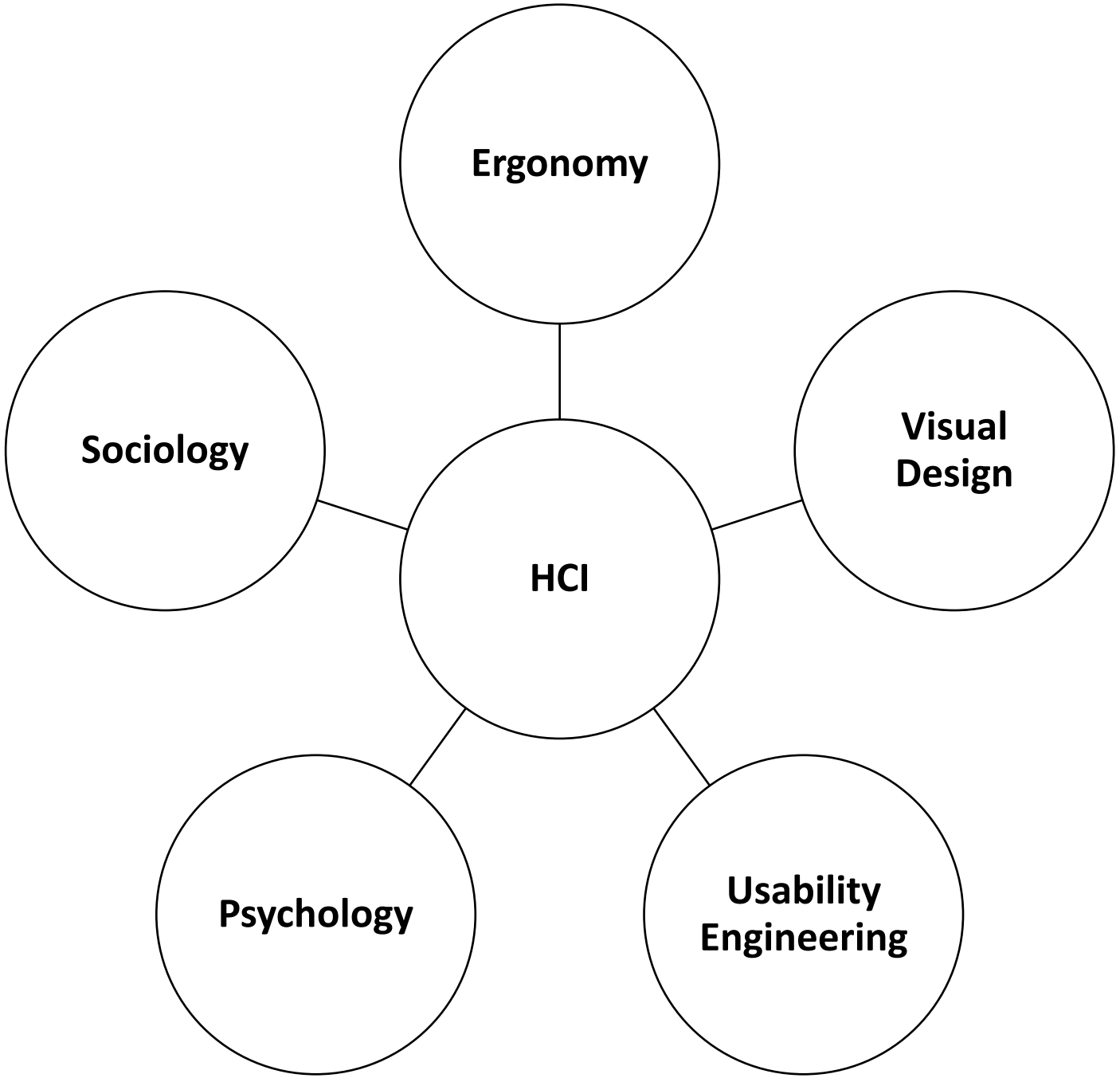
Booth outlines the Human-Computer Interaction studies with the following titles:

- What are the characteristics that affect people's use of technology?
- What are the aspects of technology that affect people's use of technology?
- How do people acquire and conceptualize their interactive abilities?
- How do we match people's needs with technical possibilities?
- How can usable technologies be designed?
- How does technology affect organizations? (Booth, 1989)

Green - 2008

In contrast to Booth's list, Green summarized the scope of Human-Computer Interaction with the following items:

- Developing processes and methodologies for designing interfaces,
- Developing techniques for evaluating and comparing interfaces,
- Developing new interface and interaction techniques,
- Developing models and theories that describe and predict interaction,
- Designing systems that will minimize the distance between the cognitive model in the process of doing a job and the way computers understand what the user wants to do (Green, 2008)


Ergonomy
Sociology
Visual Design
HCI
Psychology
Usability Engineering

History of HCI

Switch Panels

- 1950s - Hardware Level
- For engineers

Cards

- 1960s Batch processes

Fortran/Cobol

- 1970s – Programming Languages

Command Line Interface

- 1970s – Terminals
- Text based commands &

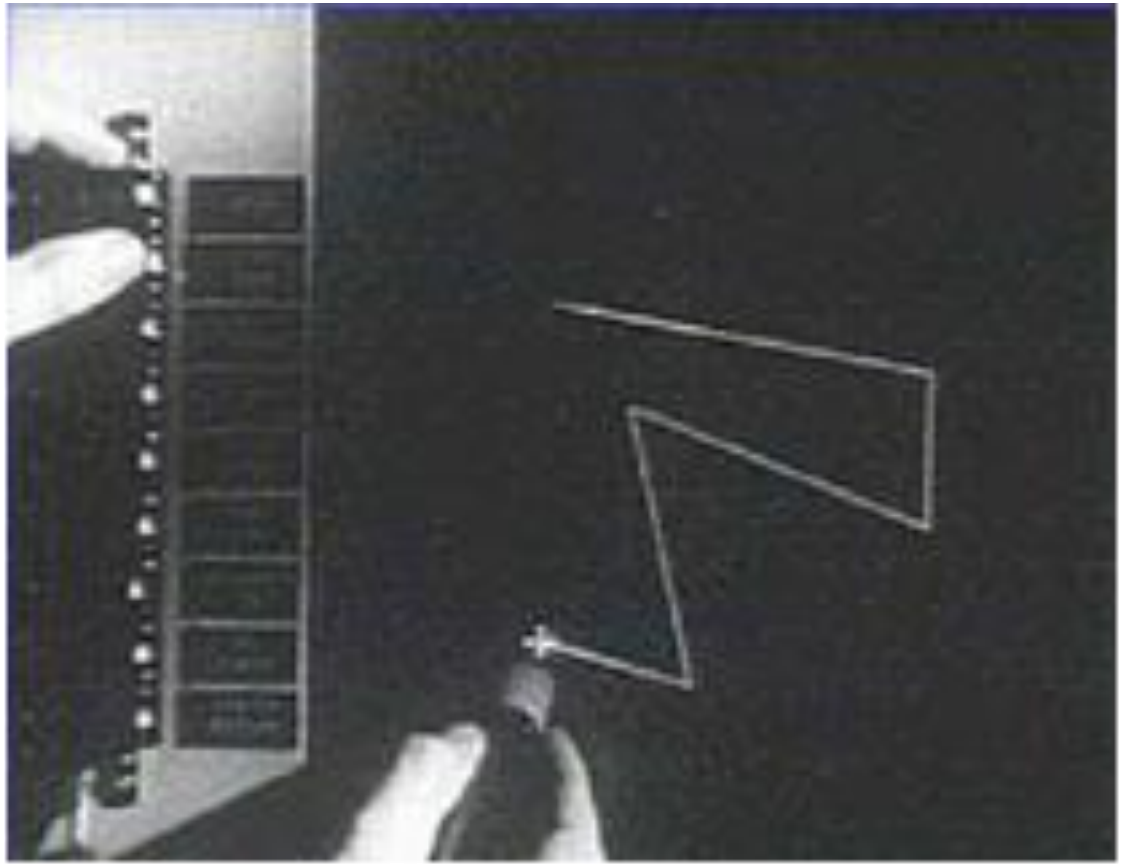
Graphical UI

- 1980s
- 2D images & Icons

Natural UI

- 1990s – intuitive inputs
- Interaction with voice, mimic, eye movement

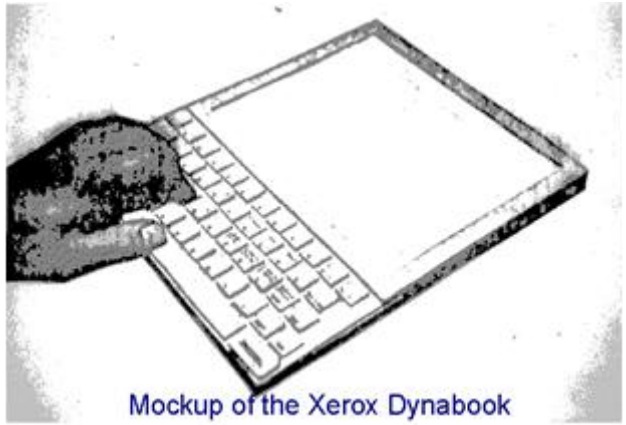
Ivan Sutherland – Sketch Pad,& Light pen 1963



Engelbart, Mouse, 1964

Engelbart, High Resolution 2d Screen, 1964

Alan C. Kay 1969 - Dynabook

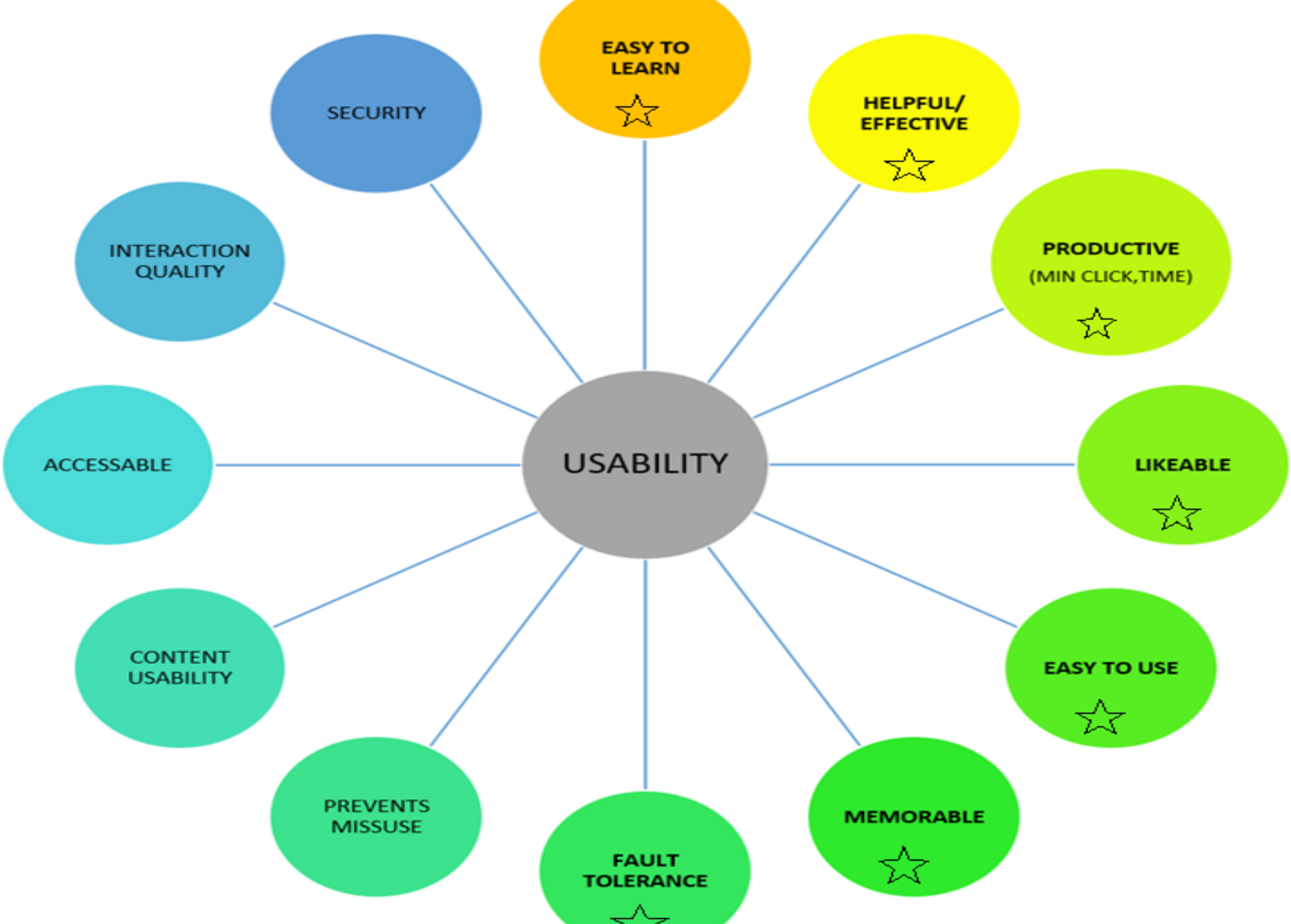


Usability

A quality element that shows how easy it is to use the interface of a product.

Systems that are not user-friendly are bound to remain idle.

Around 63% of large software projects exceed their cost estimates (Lederer, 1992) and 80% of software life-cycle costs occur during the maintenance phase because of unforeseen usability problems (Pressman, 1992).


USABILITY
EASY TO LEARN
HELPFUL/ EFFECTIVE
PRODUCTIVE (MIN CLICK,TIME)
LIKEABLE
EASY TO USE
MEMORABLE
FAULT TOLERANCE
PREVENTS MISSUSE
CONTENT USABILITY
ACCESSABLE
INTERACTION QUALITY
SECURITY

The most critical elements of usability;

- ✓ ability to achieve the purpose with the **least effort and time**,
- ✓ **easy** to use,
- ✓ easy to **remember**,
- ✓ **satisfactory** design in the first use,
- ✓ **preventing** mistakes,
- ✓ Fault tolerance (ability to take back mistakes**)**,
- ✓ a **pleasant user experience**.

Usability Criteria

Effectiveness
Meeting the requirements, fitness for purpose, content suitability, accuracy, context of use

Productivity / efficiency
Processing speed, time spent/number of clicks/number of steps to complete a task, low error rate, information organization, accessibility

Learnability
Initial performance, use of advanced features, terminology, including shortcuts, including tips, design integrity, distinctive features, help

Usability Criteria

Memorability
How hard is it to memorise the usage of the system in users' next visit.

Satisfaction
First impression / perception, ease of use, aesthetic and simple design, having a directive design that reduces computer use anxiety, readability, participation, loyalty, security, increasing the desire to learn, frequency of complaints, preferring the system to others

Error Management
Preventing errors with routing design and adequate warning messages, flexible use with the ability to revert back the operations.

Dr. Shneiderman - The 8 Golden Rules of Design

1. Design must be consistent
2. It should contain shortcuts and tips
3. It should give informational messages about the actions taken.
4. Transaction process for each transaction group must be specific and sequential
5. It should minimize the possibility of the user to make mistakes, and should be descriptive in case of error.
6. It should allow to rollback transactions
7. It should be felt that the transactions are under the control of the user.
8. Resources should be used efficiently, memory space should not be occupied too much (Shneiderman, 1987)

Donald Norman - 1988 - User-Centered Design Guidelines

1. It should be clear what operations the user can perform at any time.
2. The general structure of the system should be shown to the user, so that the user can draw his own path.
3. The user should be informed about the current status of the system.
4. The user should not be confused with designs contrary to general use. (Norman, 1988)

Norman, unlike Shneiderman, mentioned the importance of user habits in item 4 and stated that no designs should be made against the habits.

Christopher Wickens - 2004 - Design Principles

I. Perceptual Principles:

1. Items on the screen must be legible,
2. The user should not be asked questions that are open to interpretation (color, loudness, size, etc.),
3. Avoid designs contrary to the user's expectations,
4. The harmony between the forms allows the user to learn faster,
5. Confusion should be cleared by giving distinctive signals for different situations.

II. Logical Model Principles

1. The images/icons used should resemble the object they represent.
2. The movement of the moving elements must conform to the logic model of the user.

Christopher Wickens - 2004 - Design Principles

III. Principles of Attention

1. The cost of accessing information should be minimized. Ex: Number of clicks
2. By giving the information to the user through more than one channel, it can be provided to the user to perceive it more quickly. Ex: Both as a picture and text.

IV. Memory Principles:

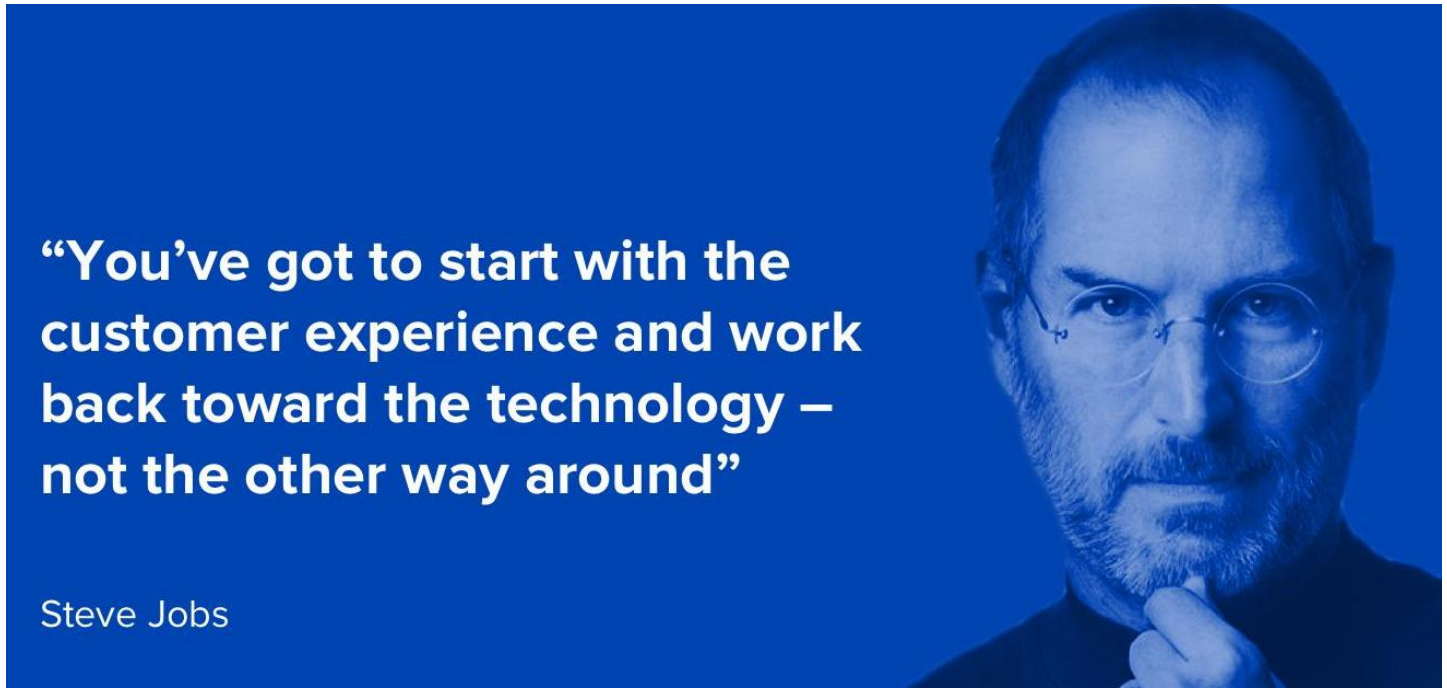
1. The user should not have to remember some information. Information should be presented to the user as options.
2. The user should be helped to plan ahead by giving some estimated information. Ex: "There are 10 minutes left to finish the process"
3. The behavior of the items on the screen should be used on similar screens to ensure harmony with the old information in the user's memory. (Wickens, Christopher D., John D. Lee, Yili Liu, and Sallie E. Gordon Becker., 2004)

What do professionals do in the HCI business?

- **interaction designers** - people involved in the design of all the interactive aspects of a product
- **usability engineers** - people who focus on evaluating products, using usability methods and principles
- **web designers** - people who develop and create the visual design of websites, such as layouts
- **information architects** - people who come up with ideas of how to plan and structure interactive products
- **user experience designers** - people who do all the above but who may also carry out field studies to inform the design of products

User Centered Design (UCD)

User-centered design (UCD) is **an iterative design process in which designers focus on the users and their needs in each phase of the design process**. In UCD, design teams involve users throughout the design process via a variety of research and design techniques, to create highly usable and accessible products for them.



User Centered Design (UCD)

User-centered design means working with your users all throughout the project.

— Donald A. Norman —

User-centered design means understanding what your users need, how they think, and how they behave - and incorporating that understanding into every aspect of your process.

— Jesse James Garrett —

User Experience
Design

What is involved in the process of User Centered Design

- Identify needs and establish requirements
  - Understand your users (needs, ‘wants’, goals, tasks, context, etc.)
- Develop alternative designs
- Build interactive prototypes that can be communicated and assessed
- Evaluate what is being built throughout the process

Improving Interfaces

- Know the User!
  - Physical abilities
  - Cognitive abilities
  - Personality differences
  - Skill differences
  - Cultural diversity
  - Motivation
  - Special needs

UCD

- Don’t assume all users are alike
- Don’t assume all users are like the designer
- Users bring vital knowledge to design:
  - They know a lot about the *problem*
  - They know a lot about the current tools
  - They typically know **very little about design**

Human information processing limitations, human cognitive and sensory limits.

Elements for interaction

1. the user’s motor system (hand, voice, etc.),
2. the user’s sensory system (visual system, tactile receptors, acoustic system,etc.),
3. the computer’s input devices (keyboard, mouse, data glove, etc.), and
4. the computer’s output devices (graphical displays, all kinds of virtual realityoutput devices, etc.)

Cognitive Researches

- Cognitive models represent users of interactive systems.
- Hierarchical models represent a user's task and goal structure.
- Linguistic models represent the user-system grammar.
- Physical and device models represent human motor skills.
- Cognitive architectures underlie all of these cognitive models.

Cognitive models

- They model aspects of user
  - understanding
  - knowledge
  - intentions
- processing
  - Common categorisation
  - Competence vs. Performance
  - Computational flavour

Know the power of Visual perception!

What happens when someone looks to your design creations?

How does brain react to the UI elements you put into your software GUI is very critical for usability?

Gestalt psychology suggests that we do not simply focus on every small component. Instead, our minds tend to perceive objects as part of **a greater whole** and as elements of more complex systems. (usertesting.com, Verywell.com)

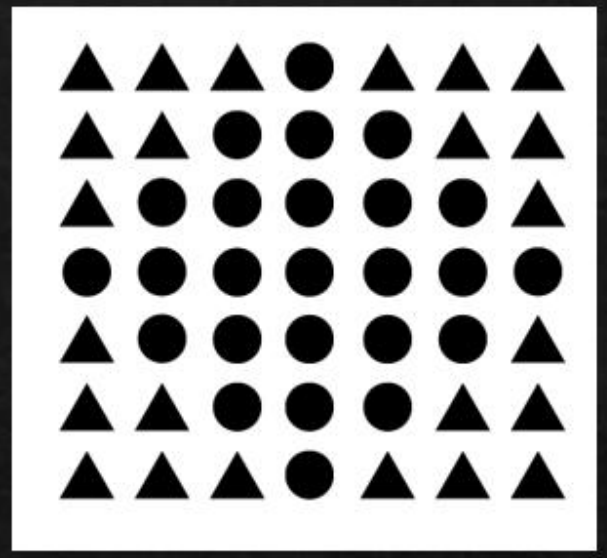
GESTALT PRINCIPLES-1

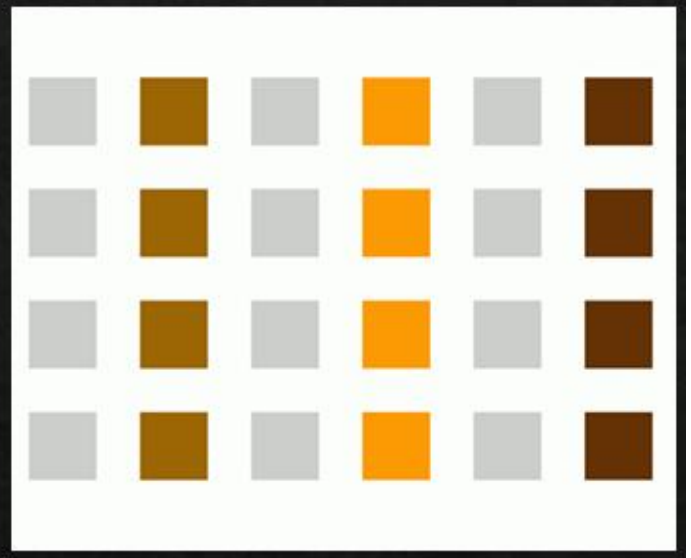
«FIGURE or GROUND»

Human brain first determines which is the figure and which is the ground in a visual. This lets us know what we should be focusing on and what we can safely ignore in a composition.

GESTALT PRINCIPLES-2

«SIMILARITY»

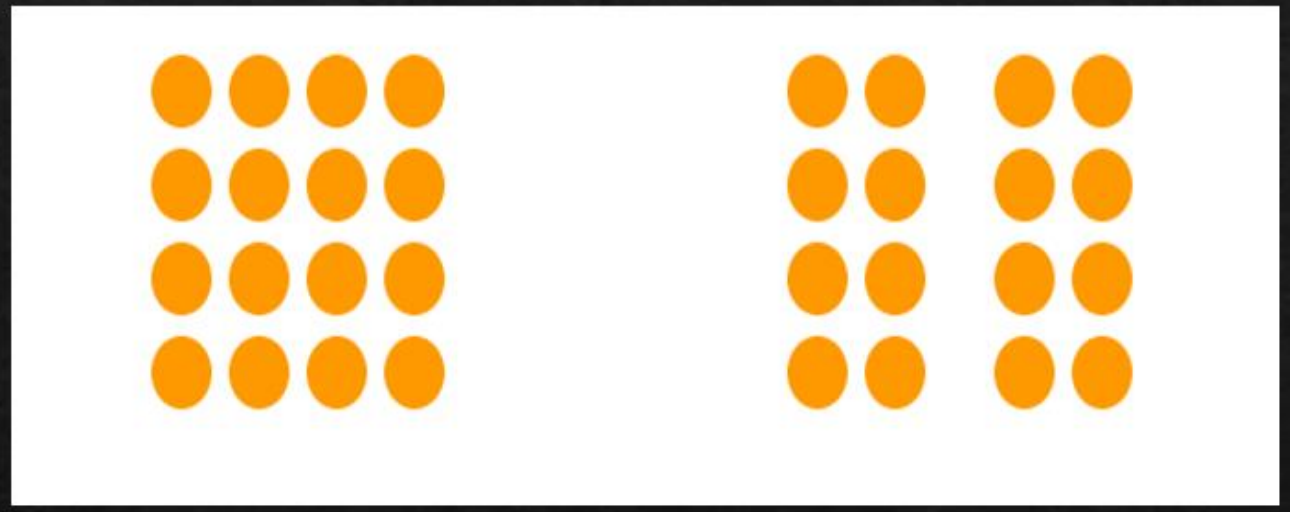




The principle of similarity states that when things appear to be similar to each other, we group them together. And we also tend to think they have the same function.

GESTALT PRINCIPLES-3

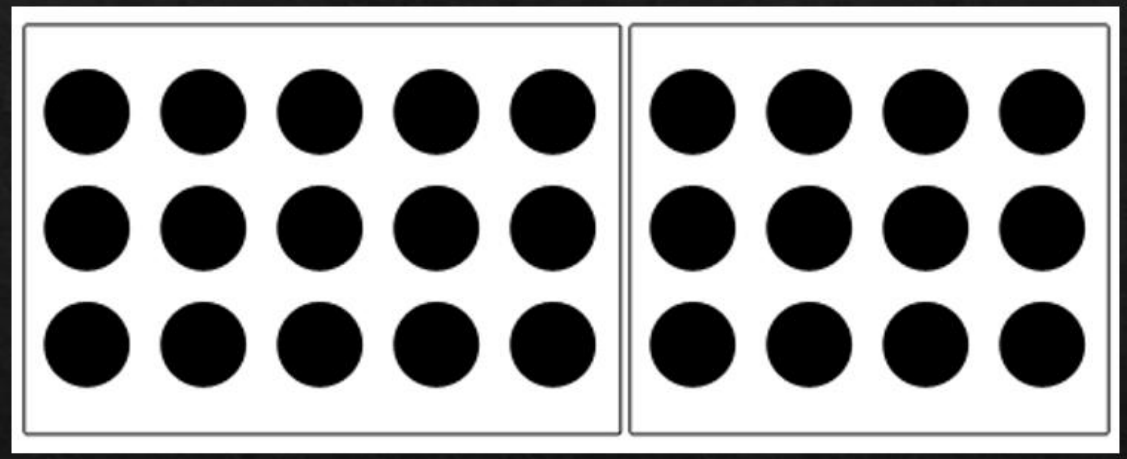
«PROXIMITY»



The principle of proximity states that things that are close together appear to be more related than things that are spaced farther apart.

GESTALT PRINCIPLES-4

«COMMON REGION»

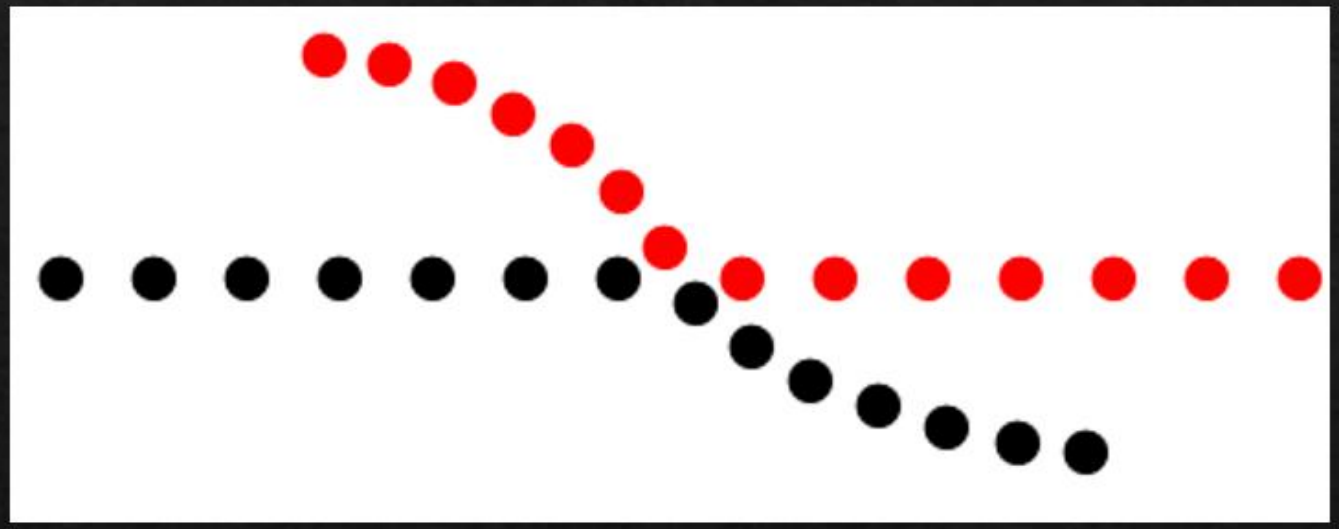


The principle of common region states that when objects are located within the same closed region, we perceive them as being grouped together.

Adding borders or other visible barriers is a great way to create a perceived separation between groups of objects–even if they have the same proximity, shape, color, etc.

GESTALT PRINCIPLES-5

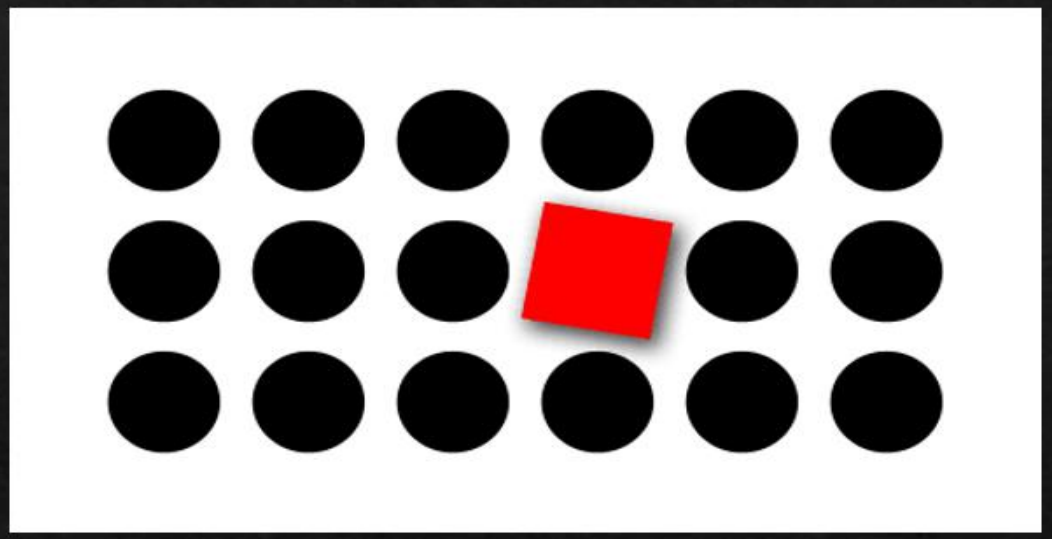
«CONTINUITY»



The principle of continuity states that elements that are arranged on a line or curve are perceived to be more related than elements not on the line or curve.

GESTALT PRINCIPLES-6

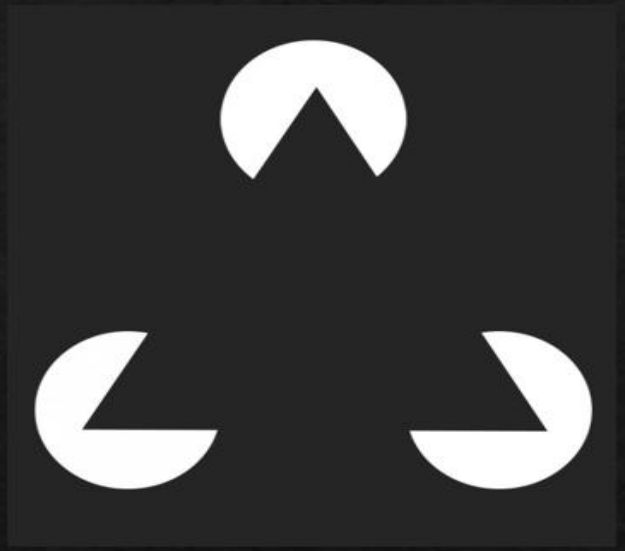
«FOCAL POINT»



The focal point principle states that whatever stands out visually will capture and hold the viewer's attention first and it's the first point of interest that grabs your attention, and from there your attention moves to other parts of the image.

GESTALT PRINCIPLES-7

«CLOSURE»



The principle of closure states that when we look at a complex arrangement of visual elements, we tend to look for a single, recognizable pattern.

In other words, when you see an image that has missing parts, your brain will fill in the blanks and make a complete image so you can still recognize the pattern.

Miller's Law - The limit of human capacity for processing information

- Most people stored 7 items +/- 2 in their short-term memory.
- Going beyond this limit made it very difficult for people to memorize information.
- This was especially important for those who were presented with the information for the first time because they haven't had the time to encode the information into long-time memory.

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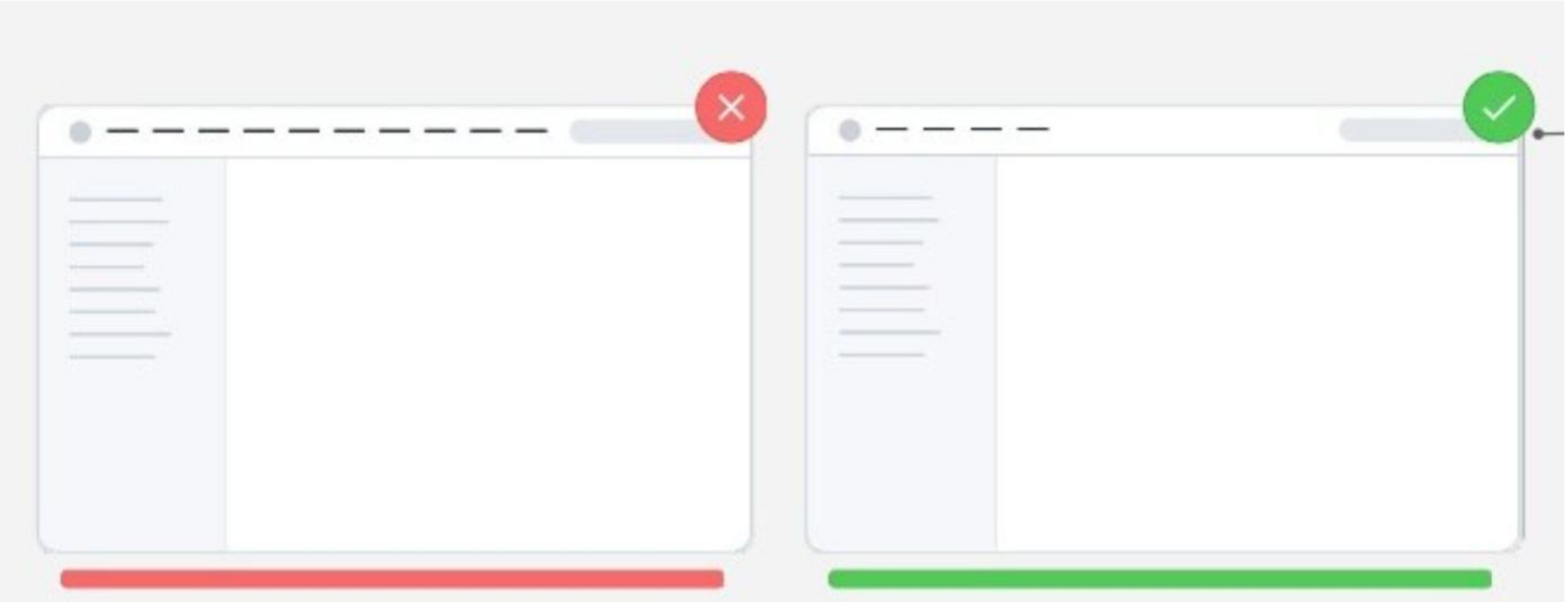
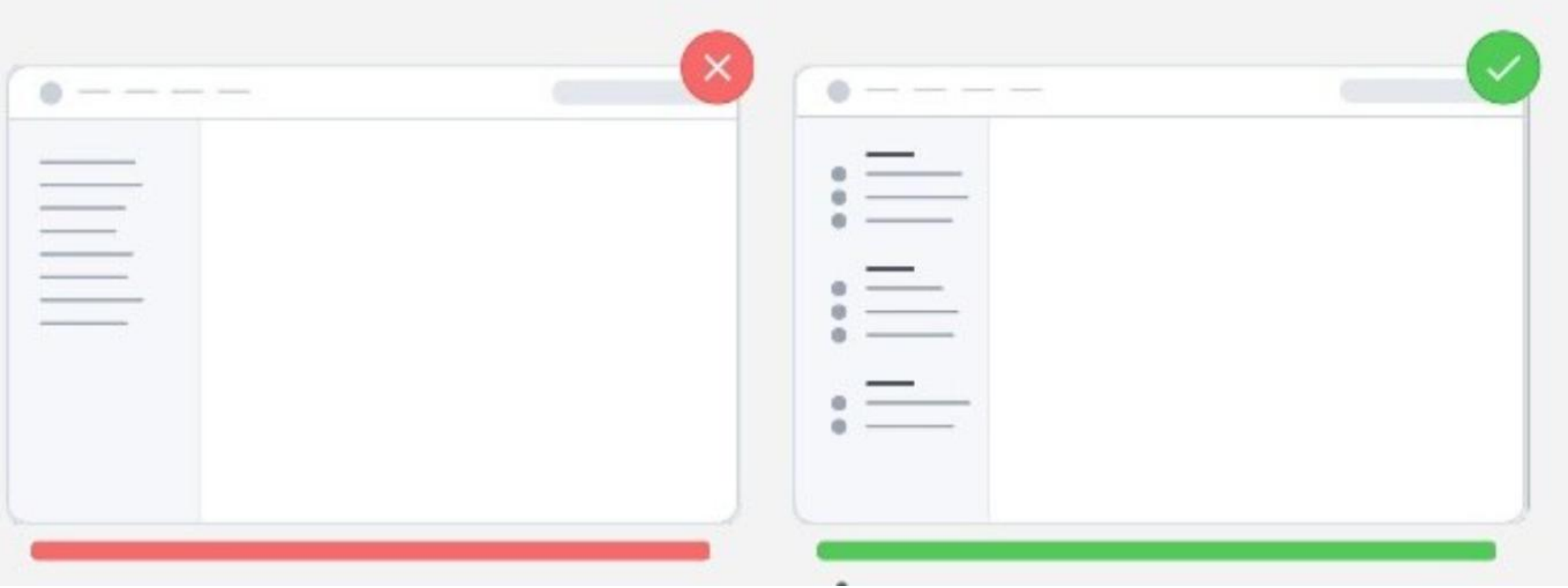
Miller’s Law



Miller’s Law



Fitt’s Law

The time it takes to reach a target is a function of the **distance** and the size of the **target**. If a target is wide and large and the distance is short, we can reach the target in a shorter time. (Fitts, 1954)

In the example below, the clickable red areas are enlarged, making it easier for the user to reach the target.

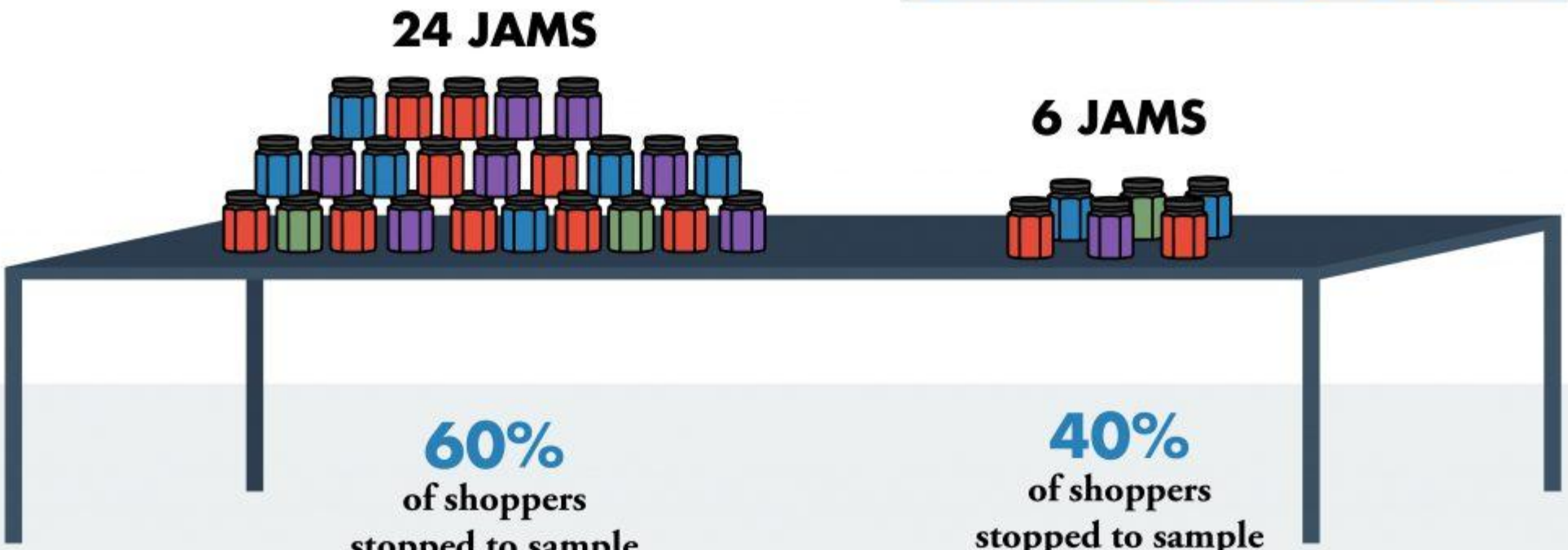
Fitt’s Law

Hick’s Law

Hick suggested that, as the number of options available to the user increases, the time to make a choice increases.

When the user is presented with more options and information, they become more confused. This rule should be considered when designing user interfaces. (Hick, 1952)

JAM TASTING STUDY



24 JAMS	6 JAMS
60% of shoppers stopped to sample	**40%** of shoppers stopped to sample
2 flavors sampled, on average	**2** flavors sampled, on average
3% of shoppers bought jam	**30%** of shoppers bought jam

Evaluating usability.

Predictive and interpretive evaluation.

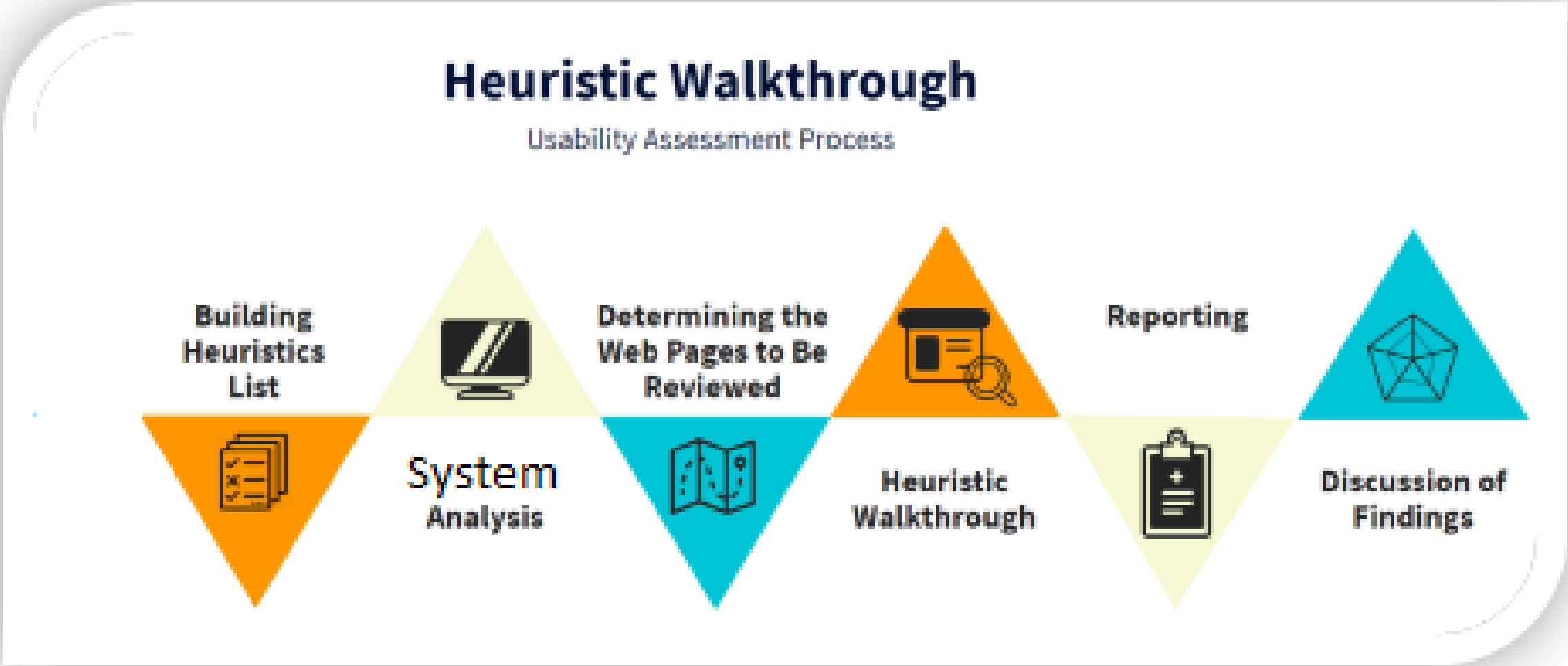
Test Methods

- Expert Analysis
- Questioning Methods
- Remote Usability Testing
  - Server Log Analysis
    - Statistical Log Analysis
    - Log Path Analysis
  - Client Log Analysis
- Formative Tests
- Eye Tracking

Expert Analysis Methods

These methods are based on one or more usability experts to examine the system in the direction of usability principles by putting themselves in the user position. Cognitive Animation and Heuristic Walkthrough are in this group.

A Heuristic Walkthrough is an expert evaluation in which one or more usability experts review the system via a heuristic criterion checklist. This process cannot be automated and also, experts should be eligible both in usability testing and in the domain of the software.


Heuristic Walkthrough
Usability Assessment Process
Building Heuristics List
System Analysis
Determining the Web Pages to Be Reviewed
Heuristic Walkthrough
Reporting
Discussion of Findings

Heuristics

In the Heuristic Walkthrough method, the researcher determines the evaluation criteria based on the time allocated for the research and the scope of the study (Nielsen, 1994).

Heuristics List

- System status visibility (remaining time to finish the background process, transaction result)
- User control and freedom (feeling lost, navigation features, etc.)
- Consistency (using the same icons for the same type of transaction)
- Avoiding errors (warning messages, preventing non-standard operations)
- Flexibility and efficiency of use (performing the transaction in the least steps, sorting in lists and presenting with filtering features, Minimizing the cost of accessing information. (Ex: Number of clicks)
- Aesthetic and simple design (avoiding unnecessary use of colour and information)
- Error Messages (clear, understandable, guiding)
- Help and documentation (to be complete and sufficient)
- Including shortcuts and tips (quick access to the latest / most frequently used pages, information notes and tips about transactions, sitemap, etc.)

Heuristics List

- Allowing Roll Back (fault tolerance)
- Design Against General Use (links are underlined texts, radio buttons are used for single selection, etc.)
- Legibility of items on the screen
- Harmony between forms and faster learning of the user (flow in forms (horizontal/vertical), implementation of a design determined in button and label placement in the whole system)
- Eliminating confusion by giving distinctive signals for different situations (active and passive buttons look different, etc.).
- The pictures and icons used to resemble the object they represent (using icons expressing the function)
- The screen elements expected to be clicked should be large enough and together (page number buttons should be large enough to be easily clicked, labels in the forms and the relevant data entry area close to each other, etc.)
- Not providing more options and information to the user than necessary (not showing authorized / out-of-duty items)

System Analysis & Determining the items to be evaluated

All menus, pages and functions in the system should be reviewed
The pages should be listed, prioritized and divided into categories below.

- Basic function
- Assistant, service
- Frequent adjustments/maintenance
- Initial setup and settings
- Periodic maintenance and maintenance

Run Heuristic Evaluation

Review the following attributes of the items regarding the heuristics, which consist of general rules that define common properties of user interfaces:

- the purpose and shape of all page elements on the pages,
- placement,
- visual features,
- labels,
- the terminology,
- the relationship of the page elements,
- the distance and sequence between the page elements,
- page flows & directions,
- the complexity of the pages,
- the use of colour,
- existence of the help documents,
- information/error/warning message
- the number of steps for each task.

Run Heuristic Evaluation

We apply Heuristic Walkthrough in two iterations on the system.
The first iteration is about getting an idea about workflow of the entire system.
In the second iteration, we focus on the interface elements deeply with the whole system knowledge(Nielsen, 1994).

Reporting

The findings and the screenshots of the problematic interfaces should be reported along with the usability checklist.

	Page 1	Page 2	Page 3	Page 4	Page n
Heuristics 1	problem …	problem …		problem …	
Heuristics 2			problem …		
Heuristics 3	problem …		problem …	problem …	
Heuristics 4		problem …	problem …		problem …
Heuristics 5	problem …	problem …			
Heuristics 6	problem …				
Heuristics 7				problem …	problem …
Heuristics 8	problem …	problem …			
Heuristics 9		problem …			
Heuristics 10		problem …		problem …	
Heuristics 11			problem …		
Heuristics 12					problem …
Heuristics 13	problem …				
Heuristics 14		problem …			
Heuristics 15					

Screenshots + videos